Assignment Part 2: Urdu Speech Dataset Banana

Sections: 7C, 7D, 7F, 7G, 7J
Har student: 500 recordings
Har recording ki length: _____ seconds
Deadline: 23 October 2026
Submission: Google Drive (link Section 13 mein hai)

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1. Is assignment mein kya karna hai?

Part 1 mein aap ne ek Urdu recording banayi thi aur us ka text **Urdu** aur **Roman Urdu** dono mein likha tha.

Ab Part 2 mein aap ne **500 recordings** ka ek **standard speech dataset** banana hai. Har recording ke saath yeh kaam karne hain:

1. **Recording level:** poori recording aur us ka poora text (Urdu + Roman Urdu).
2. **Sentence level:** recording ko alag alag jumlon (sentences) mein kaatna.
3. **Word level:** har sentence ko alag alag alfaaz (words) mein kaatna.
4. **Gender label:** har file ke saath awaaz ka gender, **M** (Male) ya **F** (Female).
5. **Train/Test split:** 400 recordings **train** mein, 100 recordings **test** mein.
6. **Experience form:** dataset banane ka apna tajurba ek chhote form mein likhna.

Kaun sa standard?

Hum **Hugging Face AudioFolder** format use kar rahe hain. Aaj kal duniya bhar mein speech datasets isi tarah share kiye jaate hain (Hugging Face dunya ka sab se bara dataset platform hai). Agar aap ka folder is format mein hoga to koi bhi shakhs aap ka dataset **Python ki ek line** se load kar sakega (Section 14 dekhein).

> **Sample dekhein:** Is assignment ke saath ek sample zip `7C_042_M.zip` di gayi hai. Sample mein sirf 5 recordings hain aur asal awaaz ki jagah "beep" sounds hain, sirf format samjhane ke liye. Aap ka dataset bilkul usi tarah ka hoga, bas 500 recordings ke saath.

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2. Apni ID (Speaker ID) banayein

Har student ki ek ID hogi. Yahi ID har file ke naam mein aayegi.

Format: `SECTION_ROLLNO_GENDER`

Hissa	Kya likhna hai	Misaal
SECTION	Apna section, capital letters mein	`7C`, `7D`, `7F`, `7G`, `7J`
ROLLNO	Roll number, **hamesha 3 digits** (shuru mein zero lagayein)	Roll no 5 → `005`, 42 → `042`, 118 → `118`
GENDER	`M` = Male, `F` = Female	`M`

Misaalein:

- Section 7C, roll no 42, larka → `7C_042_M`
- Section 7J, roll no 7, larki → `7J_007_F`

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3. Folder ka structure

Aap ki zip file ke andar bilkul yeh structure hona chahiye:

```
7C_042_M/
├── README.md          ← dataset card (aap ki maloomat)
├── EXPERIENCE.md        ← aap ka tajurba (short form)
├── recordings/
│  ├── train/
│  │  ├── metadata.csv
│  │  ├── 7C_042_M_R001.wav
│  │  ├── ...
│  │  └── 7C_042_M_R400.wav
│  └── test/
│    ├── metadata.csv
│    ├── 7C_042_M_R401.wav
│    ├── ...
│    └── 7C_042_M_R500.wav
├── sentences/
│  ├── train/
│  │  ├── metadata.csv
│  │  ├── 7C_042_M_R001_S01.wav
│  │  └── ...
│  └── test/
│    ├── metadata.csv
│    └── ...
└── words/
  ├── train/
  │  ├── metadata.csv
  │  ├── 7C_042_M_R001_S01_W01.wav
  │  └── ...
  └── test/
    ├── metadata.csv
    └── ...
```

Yaani **3 levels** (recordings, sentences, words), har level mein **2 splits** (train, test), aur har split mein ek `metadata.csv` aur us ki `.wav` files.

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4. File naming ke rules

File	Naam ka format	Misaal
Recording	`ID_R###.wav`	`7C_042_M_R001.wav`
Sentence	`ID_R###_S##.wav`	`7C_042_M_R001_S01.wav`
Word	`ID_R###_S##_W##.wav`	`7C_042_M_R001_S01_W03.wav`
Zip file	`ID.zip`	`7C_042_M.zip`

- `R001` se `R500` tak recordings (**3 digits**).
- `S01`, `S02` ... har recording mein sentences dobara `S01` se shuru.
- `W01`, `W02` ... har sentence mein words dobara `W01` se shuru.
- `7C_042_M_R237_S02_W05.wav` ka matlab: **recording 237, doosra sentence, paanchwan word**.

Zaroori baatein:

- Naam mein **space bilkul nahi** (sirf underscore `_`).
- Digits poore likhein: `R007` (na ke `R7`), `S01` (na ke `S1`), `W05` (na ke `W5`).
- `M`, `F`, `R`, `S`, `W` aur section hamesha **capital** mein. Extension `.wav` chhote letters mein.

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5. Train aur Test split

Machine learning mein model ko **train** data se sikhaya jaata hai aur **test** data se check kiya jaata hai. Test data model ne pehle kabhi nahi dekha hona chahiye.

Split	Recordings	Kitni
train	`R001` se `R400`	400 (80%)
test	`R401` se `R500`	100 (20%)

> **Sab se zaroori rule:** Jis recording ki file `train` mein hai, **us ke saare sentences aur words bhi `train` mein** jayenge. Isi tarah test. Misaal: `R450` test mein hai, to `7C_042_M_R450_S03_W02.wav` bhi `words/test/` mein jayegi.

Agar aik hi recording ke kuch hisse train aur kuch test mein chale gaye to test ka matlab khatam ho jaata hai (isay **data leakage** kehte hain).

Mashwara: Test recordings (R401–R500) mein woh jumlay na dohrayein jo train recordings mein bol chuke hain.

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6. Audio ki settings

Har `.wav` file in settings mein honi chahiye:

Setting	Value
Format	WAV
Sample rate	**16000 Hz**
Channels	**Mono** (1 channel)
Bit depth	**16-bit PCM**

Recording aur kaatne ke waqt dhyan rakhein:

- Khamosh kamre mein record karein (pankha, TV, traffic ki awaaz kam se kam).
- Har recording mein mukhtalif jumlay bolein. 500 recordings mein ek hi jumla baar baar na dohrayein.
- Word ka koi hissa kata hua na ho (shuru ya aakhir ki awaaz poori ho).
- Sentence file ke shuru aur aakhir mein thori khamoshi (0.1 se 0.3 second) theek hai. Word files mein faltu khamoshi kam se kam.

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7. Text likhne ke rules

Urdu aur Roman Urdu dono

- Jo bola gaya hai **bilkul wohi** likhein. Apni taraf se kuch na badlein.
- Agar "uhh", "umm" bola ya koi lafz atak gaya to **woh recording dobara karein**.
- Numbers ko alfaaz mein likhein: `20` nahi, **بیس / bees**.

Sabse zaroori rule: words ki ginti barabar honi chahiye

Har sentence mein **Urdu words ki ginti = Roman Urdu words ki ginti**, taake har word ka joda ban sake.

	Urdu	Roman Urdu	Words
✓ Sahi	آج موسم بہت اچھا ہے	Aaj mausam bohat acha hai	5 = 5
× Ghalat	آج موسم بہت اچھا ہے	Aaj mausam bohatacha hai	5 ≠ 4

Roman Urdu

- Ek lafz ki **ek hi spelling** poore dataset mein. Agar `bohat` likha to har jagah `bohat` (kabhi `bohot`, kabhi `bahut` nahi). Apni ek chhoti si spelling list bana lein.
- Sirf English letters (a-z). Koi special sign ya accent nahi.

Urdu

- Urdu keyboard se likhein (Gboard ya Windows Urdu keyboard). Hindi ya Arabic keyboard nahi.

Punctuation

- Recording aur sentence ke text mein aakhir mein `۔` (Urdu) aur `.` (Roman) laga sakte hain.
- **Word ke text mein koi punctuation nahi.**

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8. Audio kaise kaatein (Audacity, free software)

Audacity download karein: **audacityteam.org**

Har recording ke liye yeh steps karein. Pehli 5–10 recordings mein waqt lagega, phir haath bait jayega.

Step A: Recording tayyar karein

1. Recording Audacity mein kholein.
2. Agar stereo hai: **Tracks → Mix → Mix Stereo Down to Mono**.
3. Neeche bayein taraf **Project Rate** ko **16000** karein.
4. **File → Export → Export as WAV** (Signed 16-bit PCM), naam `7C_042_M_R001.wav`, sahi folder (`recordings/train` ya `recordings/test`) mein.

Step B: Sentences kaatna

1. Pehla sentence mouse se select karein (play kar ke check karein).
2. **Ctrl + B** dabayein aur label likhein: `7C_042_M_R001_S01`
3. Har sentence ke liye yahi karein: `..._S02`, `..._S03` ...
4. **File → Export → Export Multiple** (naye version mein: **Export Audio → Multiple Files**). Format **WAV, Signed 16-bit PCM**, "Split files based on: **Labels**", "Name files: **Using Label/Track Name**". Folder: `sentences/train` (ya `test`).
5. **File → Export → Export Labels** se `R001_sentences.txt` save karein. Is mein har sentence ka start aur end time hai, `metadata.csv` ke liye.

Step C: Words kaatna (usi recording mein)

1. Sentence wala label track hata dein (track ke **X** par click). Times aap ne Step B mein save kar liye hain. Zarurat ho to **File → Import → Labels** se wapas la sakte hain.
2. Har word select kar ke **Ctrl + B** se label lagayein: `7C_042_M_R001_S01_W01`, `7C_042_M_R001_S01_W02` ... phir S02 ke words `..._S02_W01` ...
3. **Export Multiple** se `words/train` (ya `test`) mein export karein.
4. **Export Labels** se `R001_words.txt` save karein.

> **Tip:** Zoom: **Ctrl + 1** (in), **Ctrl + 3** (out). Waveform dekh kar word ki boundary aasani se milti hai.
> **Note:** Label `.txt` files apne paas rakhein, zip mein na daalein.

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9. metadata.csv kaise banayein

`metadata.csv` ek table hai jis mein har audio file ki ek row hoti hai. Har level aur har split ki apni `metadata.csv` hai (kul 6 files).

Isay **Google Sheets** mein banayein: har level ke liye ek sheet. Phir har split ke liye **File → Download → Comma-separated values (.csv)** karein aur naam `metadata.csv` rakhein. (Google Sheets Urdu text theek save karta hai, Excel aksar kharab kar deta hai.)

Rules:

- **Pehla column hamesha** `file_name`, aur column ke naam bilkul neeche jaise (chhote letters, underscore).
- `file_name` mein sirf file ka naam, folder ka naam nahi.
- Time seconds mein, **2 decimal** tak: `3.82`.
- Saare times (sentence aur word dono) **recording file** ke hisaab se hain. Yeh wohi times hain jo Audacity ke **Export Labels**

se milte hain.

- `duration_sec` = `end_sec − start_sec`.
- Google Sheets mein `roll_no` column ko **Format → Number → Plain text** karein, warna `042` ka `42` ban jayega.

recordings/.../metadata.csv

Column	Matlab	Misaal
file_name	Audio file ka naam	`7C_042_M_R001.wav`
speaker_id	Aap ki ID	`7C_042_M`
section	Section	`7C`
roll_no	Roll no (3 digits)	`042`
gender	M ya F	`M`
recording_id	Recording number	`R001`
duration_sec	Recording ki lambai (seconds)	`7.89`
urdu_text	Poori recording ka Urdu text (ek line mein)	`آج موسم بہت اچھا ہے۔ میں روزانہ…`
roman_urdu_text	Poori recording ka Roman Urdu text	`Aaj mausam bohat acha hai. Main rozana…`

sentences/.../metadata.csv

Column	Matlab	Misaal
file_name	Audio file ka naam	`7C_042_M_R001_S01.wav`
speaker_id, section, roll_no, gender, recording_id	Upar jaise	`7C_042_M`, `7C`, `042`, `M`, `R001`
sentence_id	Sentence number	`S01`
start_sec	Recording mein sentence kahan shuru hua	`0.40`
end_sec	Recording mein sentence kahan khatam hua	`3.33`
duration_sec	end_sec − start_sec	`2.93`
urdu_text	Sentence ka Urdu text	`آج موسم بہت اچھا ہے`
roman_urdu_text	Sentence ka Roman Urdu text	`Aaj mausam bohat acha hai`

words/.../metadata.csv

Column	Matlab	Misaal
file_name	Audio file ka naam	`7C_042_M_R001_S01_W02.wav`
speaker_id, section, roll_no, gender, recording_id	Upar jaise	`7C_042_M`, `7C`, `042`, `M`, `R001`
sentence_id	Kis sentence ka word hai	`S01`
word_id	Word number	`W02`
start_sec	Recording mein word kahan shuru hua	`1.08`
end_sec	Recording mein word kahan khatam hua	`1.58`
duration_sec	end_sec − start_sec	`0.50`
urdu_word	Urdu word	`موسم`
roman_urdu_word	Roman Urdu word	`mausam`

> Sample zip ki `metadata.csv` files Google Sheets mein khol kar dekhein. Apni files bilkul waisi hi banayein.

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10. README.md (Dataset Card)

Har asal dataset ke saath ek **dataset card** hota hai jo batata hai ke dataset mein kya hai. Hugging Face par yeh `README.md` file

hoti hai.

Sample zip ki `README.md` copy karein (Notepad mein kholein) aur sirf yeh cheezein badlein:

- **Upar ka hissa (`---` ke darmiyan):** sirf `pretty_name` mein apni ID. Baaqi bilkul waisa hi rehne dein. Yeh hissa computer ko batata hai ke dataset ke 3 hissay (`recordings` , `sentences` , `words`) kahan hain.
- **Speaker table:** apna naam, section, roll no, gender, phone ka model, jagah, tareekhein.
- **Counts table:** har level aur split mein kitni files hain.

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11. Experience Form (EXPERIENCE.md)

Dataset mukammal hone ke baad apna tajurba ek chhote form mein likhein. Asal research mein bhi dataset banane wale apne masail aur dataset ki kamiyan likhte hain, taake use karne wale ko pata ho.

- File ka naam: `EXPERIENCE.md` , `README.md` ke saath (folder ke andar sab se upar).
- Sample zip mein bhara hua misaali form hai. Neeche wala khali form copy kar ke Notepad mein bharein.
- Har sawal ka jawab 1–3 lines. Koi sawal khali na chhorein.

```
# Experience Form - 7C_042_M

> Har sawal ka jawab 1-3 lines mein, Roman Urdu ya English mein. Koi jawab ghalat nahi hota, bas sach aur poora likhein.

**Naam:**
**Speaker ID:** 7C_042_M

## 1. Kul kitna waqt laga? (ghanton mein, andaza)

- Recording:
- Audio kaatna (sentences + words):
- Text likhna aur metadata banana:

## 2. Mushkil kitni lagi? (1 = bohat aasan, 5 = bohat mushkil)



## 3. Sab se mushkil hissa kaun sa tha aur kyun?



## 4. Kaun si cheez ya tareeqa sab se zyada kaam aaya?



## 5. Aap ke dataset mein koi kami ya masla? (jaise background noise, kuch words saaf nahi, spelling mein farq)



## 6. Agar dobara banana ho to kya mukhtalif karenge?



## 7. Is assignment se aap ne kya seekha?
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12. Apna dataset check karein (checker script)

Is assignment ke saath `check_dataset.py` di gayi hai. Yeh aap ka poora dataset check kar ke ghaltiyan batati hai: file naam, audio settings, missing files, metadata, words ki ginti, train/test split, aur experience form.

1. Python install karein (**python.org**). Koi aur package nahi chahiye.
2. `check_dataset.py` aur apni zip ek hi folder mein rakhein.
3. Us folder mein Command Prompt/Terminal khol kar likhein:

```
python check_dataset.py 7C_042_M.zip
```

Jab tak **"OK - koi ghalti nahi mili"** na aaye, ghaltiyan theek karte rahein. Teacher bhi submission isi script se check karenge.

> Sample zip check karni ho to: `python check_dataset.py 7C_042_M.zip --expect 5` (sample mein sirf 5 recordings hain).

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13. Zip banayein aur Google Drive par upload karein

1. Apne `7C_042_M` folder par right-click → **Send to → Compressed (zipped) folder** (Mac: **Compress**).
2. Zip ka naam aap ki ID: `7C_042_M.zip`. Zip ke andar seedha `7C_042_M` folder ho.
3. Yeh link kholein:
   https://drive.google.com/drive/u/0/folders/1Jk_wubRvMa-L9528sT0YnsqWLOVoL6IT
4. Apne **section ke folder** mein jayein (`7C` , `7D` , `7F` , `7G` ya `7J`).
5. **New → File upload** se zip upload karein. Zip bari hogi (GBs mein), is liye WiFi par upload karein aur upload poora hone ka intezar karein.

> Ghalat section ke folder mein upload ki gayi file check nahi hogi. Dobara upload karni ho to purani file delete kar ke nayi upload karein.

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14. Bonus: apna dataset Python mein load karein

Yeh dekhne ke liye ke aap ka dataset sach mein "standard" hai, Google Colab (**colab.research.google.com**) ya apne computer par:

```
!pip install "datasets[audio]<4"  # Colab mein; computer par ! ke baghair

from datasets import load_dataset

words = load_dataset("7C_042_M", "words")   # "recordings" ya "sentences" bhi
print(words)        # train: ... rows, test: ... rows
print(words["train"][0])    # pehla word: audio + text + gender
```

Agar yeh chal gaya to aap ka dataset bilkul usi tarah load ho raha hai jaise duniya ke bare speech datasets.

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15. Submit karne se pehle checklist

- [ ] `check_dataset.py` ne **"OK - koi ghalti nahi mili"** kaha.
- [ ] Zip ka naam meri ID hai (`7C_042_M.zip`) aur sahi section ke folder mein hai.
- [ ] 500 recordings: R001–R400 `train` mein, R401–R500 `test` mein.
- [ ] Har recording ke sentences aur words usi split mein hain jis mein recording hai.
- [ ] Har audio WAV, 16000 Hz, mono, 16-bit.
- [ ] Har sentence mein Urdu words ki ginti = Roman Urdu words ki ginti.
- [ ] Kuch word files sun kar check kiya ke sahi word hai aur kata hua nahi.

- [ ] Roman Urdu spelling poore dataset mein ek jaisi hai.
- [ ] README.md mein apni maloomat aur counts likhe hain.
- [ ] EXPERIENCE.md ke saare 7 sawalon ke jawab likhe hain.

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16. Aam ghaltiyan (in se bachein)

Ghalti	Sahi tareeqa
`7c_42_m_r1_s1.wav`	`7C_042_M_R001_S01.wav`
`7C 042 M R001.wav` (space)	`7C_042_M_R001.wav`
R450 ke words `words/train` mein	`words/test` mein (recording ke split ke mutabiq)
CSV mein `042` ka `42` ban gaya	Column ko Plain text format karein
Excel mein Urdu `????` ban gayi	Google Sheets se CSV download karein
`metadata.csv.csv` ya `README.md.txt`	File extensions show kar ke faltu extension hata dein
Stereo ya 44100 Hz audio	Mono aur 16000 Hz mein convert karein
Word file mein do words	Har word ki alag file
MP3 ka naam badal kar `.wav` kar diya	Audacity se asal WAV export karein

Koi sawal ho to class mein ya email par poochein. Best of luck!